

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
t _{SETTLING}	Settling time (full scale: for a 12-bit code transition between the lowest and the highest input codes when DAC_OUT reaches the final value of ±0.5LSB, ±1LSB, ±2LSB, ±4LSB, ±8LSB)	Normal mode DAC output buffer ON C _L ≤ 50 pF, R _L ≥ 5 kW	±0.5 LSB	-	1.7	3	μs
			±1 LSB	-	1.66	2.87	
			±2 LSB	-	1.65	2.84	
			±4 LSB	-	1.63	2.78	
			±8 LSB	-	1.61	2.7	
		Normal mode, DAC output buffer OFF, ±1 LSB C _L = 10 pF		-	1.7	2	
t _{WAKEUP}	Wakeup time from off state (setting the Enx bit in the DAC Control register) until the ±1LSB final value	Normal mode, DAC output buffer ON, C _L ≤ 50 pF, R _L = 5 Ω	-	5	7.5	μs	
		Normal mode, DAC output buffer OFF, C _L ≤ 10 pF	-	2	5.0		
PSRR	DC V _{DDA} supply rejection ratio	Normal mode DAC output buffer ON C _L ≤ 50 pF, R _L = 5 kW		-	-80	-28	dB
t _{SAMP}	Sampling time in Sample and Hold mode C _{SH} =100nF (Code transition between the lowest input code and the highest input code when DACOUT reaches final value +/- 1LSB)	MODE<2:0>_V12 = 100/101 (BUFFER ON)		-	0.7	2.6	ms
		MODE<2:0>_V12 = 110 (BUFFER OFF)		-	11.5	18.7	
		MODE<2:0>_V12=111 BUFFER OFF (DAC_OUT pin not connected, internal connection only)		-	0.3	0.6	μs
I _{leak}	Output leakage current	-		-	-	-	nA
C _{Iint}	Internal sample and hold capacitor	-		1.43	1.75	2	pF
t _{TRIM}	Middle code offset trim time	DAC output buffer ON		50	-	-	μs
V _{offset}	Middle code offset for 1 trim code step	V _{REF+} = 3.6 V		-	850	-	μV
I _{DDA} (DAC)	DAC quiescent consumption from V _{DDA}	DAC output buffer ON	No load, middle code (0x800)	-	315	-	μA
			No load, worst code (0xF1C)	-	450	-	
		DAC output buffer OFF	No load, middle/worst code (0x800)	-	20	-	
		Sample and Hold mode, C _{SH} =100 nF		-	315* _{Ton} / (_{Ton} + _{Toff})	-	
I _{DDV} (DAC)	DAC consumption from V _{REF+}	DAC output buffer ON	No load, middle code (0x800)	-	185	-	μA
			No load, worst code (0xF1C)	-	185	-	
		DAC output buffer OFF	No load, middle/worst code (0x800)	-	155	-	
		Sample and Hold mode, Buffer ON, C _{SH} =100 nF (worst code)		-	185* _{Ton} / (_{Ton} + _{Toff})	-	
		Sample and Hold mode, Buffer OFF, C _{SH} =100 nF (worst code)		-	155* _{Ton} / (_{Ton} + _{Toff})	-	